Reward engineering, simulator fidelity and lap-time benchmarks

RL driving agent in OCD: Barcelona and Shanghai

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2 November 2026

Reward: from line tracking to time-optimal

Old (`ref_tracking`), per 50 Hz tick, e = offset to the reference line ξ_{ref}:

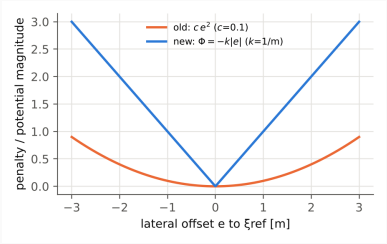
$$r = a\,\Delta s - b\,\Delta t\,\big(1 + \underbrace{\dot{u}^{\top} W \dot{u}}_{\text{jitter}}\big) - \underbrace{c\,e^{2}}_{\text{tracking}}, \qquad P_{\text{fail}} = 50 + 30\,c\,e_{\text{exit}}^{2}$$

New (`time_pbrs`, R1):

$$r = a\,\Delta s - b\,\Delta t + \big[\gamma\,\Phi(s') - \Phi(s)\big], \qquad \Phi(s) = -k\,|e(s)|,\ k = 1\,\text{m}^{-1}, \qquad P_{\text{fail}} = 100$$

1. **Jitter term removed** (logged only): top racing agents (Linesight, GT Sophy) use none.
2. **Linear $|e|$, not e^2**: constant gradient instead of flat-then-explosive.
3. **Potential-based shaping** (Ng 1999): cannot change the optimal driver; ξ_{ref} only guides early learning.

Old problem: tracking cost (175.9) exceeded lap time (123.3); the driver stayed glued to a *safe* line.



Effect of the new reward (Shanghai)

Run	Car / reward	Best clean lap 2	Own QSS (safe line)
full1	AV-21, old reward	122.85 s	133.9 s
r1_warm	AV-21, new reward, warm start	123.04 s	133.9 s
sf70h_mac_r1	sf70h_tum, new reward, from scratch	**103.24 s**	104.78 s
p1_TUM	sf70h_tum + TUM-like setup, new reward	**101.27 s**	103.56 s

- The warm start kept the old line-following habits and gained nothing.
- From scratch, the free driver **beats the QSS lap on its own safe line** by 1.5–2.3 s, because it is no longer tied to that line.
- Cost: a lower clean rate (48/194 clean deterministic evaluations in sf70h_mac_r1).

Vehicle dynamics: three cars, built in layers

	`av21` (shipped OCD)	`sf70h_tum` (27 Oct)	`sf70h_f1` (1 Nov)
Idea	Dallara AV-21 (Indy Autonomous Challenge)	AV-21 model with every value TUM's `F1_Shanghai.ini` gives	`sf70h_tum` + F1 values from papers and the FIA rules
Mass, geometry, aero	AV-21 (800 kg, 3.2 m)	TUM (733 kg, 3.6 m)	TUM
Tyre peak μ	AV-21 (1.46 / 1.16)	TUM (1.65–2.15)	TUM
Tyre slip stiffness, shape	AV-21	AV-21	PL14 (Limebeer)
Inertias	AV-21	scaled ($I \propto mL^2$)	scaled
Steering lock	0.6 rad	0.6 rad	0.30 rad
Heave springs	off	off	1.0 / 0.25 MN/m
Brakes	12 kN m, 0.50	12–15 kN m, 0.50	15 kN m, 0.58
Footprint	4.2×2.0 m	4.6×2.0 m	5.006×2.07 m

Each car is opt-in; `av21` and `sf70h_tum` stay bit-identical (85 unit tests, Shanghai `lap.csv` sha256 unchanged).

sf70h_tum: what was copied from TUM

	AV-21	sf70h_tum
Mass	800 kg	733 kg
Wheelbase	3.2 m	3.6 m
CoG height	0.30 m	0.335 m
Track F / R	1.6 / 1.5 m	1.6 / 1.6 m
Rolling radius	0.30 m	0.3299 m
I_z [kg m^2] (scaled)	1000	1159.6

Also new: ERS (120 kW, 4 MJ/lap, no recuperation), DRS (QSS only), a friction *ellipse* in the QSS, brakes 12 → 15 kN m (14.3 needed to lock at 90 m/s). RL runs use ERS/DRS off.

Tyre peak friction, mapped exactly. Both laws are linear in load with $F_{\text{nom}} = 3$ kN:

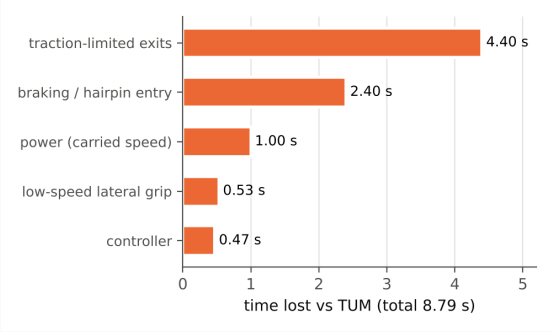
$$\mu_0 + \tfrac{d\mu}{dF_z}(F_z - f_{z0}) \;=\; (P_{D1} + P_{D2}\, df_z)\,\lambda_\mu$$

$\Rightarrow P_{D1} = \mu_0/\lambda_\mu$, $P_{D2} = \tfrac{d\mu}{dF_z} f_{z0}/\lambda_\mu$:

	μ_0	P_{D1} new	AV-21
Front x/y	1.65/1.85	1.701/1.907	1.5/1.2
Rear x/y	1.95/2.15	2.010/2.216	1.5/1.2

Grip alone: QSS 126.4 → 106.4 s. Only these 4 numbers per wheel changed; slip stiffness and shape stayed AV-21.

Diagnosis: TUM's 94.9 s lap driven closed-loop in OCD



- TUM-aligned debug car; a 50 Hz controller tracks TUM's line and speed (feed-forward + PI, traction control, ABS).
- Driven **102.99 s** vs our QSS 95.73 s vs TUM 94.89 s. The *quasi-static* limits match; the time is lost in *transients* (corner exits, braking).
- Longitudinal slip stiffness $\times 1.6$ alone: -2.69 **s**, QSS unchanged.

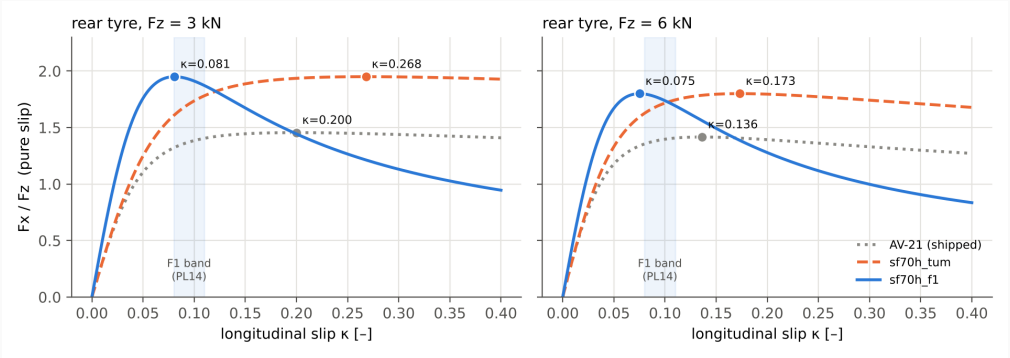
Why the AV-21 slip stiffness hurts, and the F1 fix

$$F_x = D\sin\big(C\arctan(B\kappa - E(B\kappa - \arctan B\kappa))\big), \quad D = \mu F_z, \quad B = \frac{K_x}{CD} \;\Rightarrow\; \kappa^* \propto \frac{\mu}{K_x/F_z}$$

- `sf70h_tum` raised μ by 34–79 % and kept the AV-21 stiffness K_x (25–33 % softer than F1 at low load), so the force peak **moved out to large slip**.
- **Fix (option B, approved 1 Nov):** the PL14 tyre $\mu\sin(Q\arctan(S\rho))$ is MF with $C = Q = 1.9$, $E = 0$, so $K_x/F_z = 2.747\,\mu_{\mathrm{TUM}}(F_z)/\kappa_{\max}(F_z)$, with $\kappa_{\max}$ 0.11 (2 kN) → 0.10 (6 kN). Fitted: front PKX1/2 = 42.17 / −0.96, rear 49.84 / −0.38 (AV-21 30.7 / 0.27); PCX1 = PCY1 = 1.9; all PE$*$ = 0. Peak μ unchanged.
- Cornering stiffness (PKY) kept: the AV-21 K_y is within 4 % of PL14.

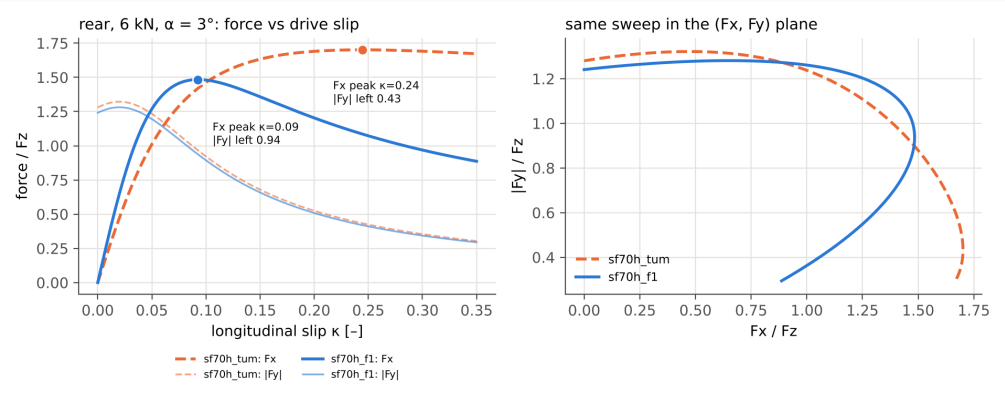
Rear tyre, peak slip	AV-21	`sf70h_tum`	`sf70h_f1`
κ^* at 3 / 6 / 12 kN	0.20 / 0.14 / –	0.27 / 0.17 / 0.10	0.081 / 0.075 / 0.064
α^* at 3 / 6 / 12 kN	5.8 / 4.4 / – °	10.8 / 8.2 / 7.8 °	8.2 / 8.6 / 10.8 °

Tyre force curves: where the peak is



Same peak μ, but the F1 tyre reaches it at $\kappa \approx 0.08$ (F1 band 0.08–0.11); `sf70h_tum` needs 0.17–0.27. Curves from a Python copy of OCD's MF5.2 that reproduces the committed sweep exactly.

Combined slip: what a corner exit looks like



At $\alpha = 3°$, `sf70h_tum` reaches its Fx peak only at $\kappa = 0.24$, with 0.43 of lateral force left; the F1 tyre peaks at $\kappa = 0.09$ with 0.94 left. Caveat: the combined-slip weights are still AV-21 in both cars, so lateral force falls ≈30 % by $\kappa = 0.1$, which is harsher than TUM's ellipse.

sf70h_f1: other changes, and a new finding

- **Steering lock** 0.6 → 0.30 rad (F1 normal 17°). The OCP needs at most 0.27 rad.
- **Heave springs** $k = m_a(2\pi f)^2$: front 1.0 MN/m (12.95 Hz, 9× the corner spring), rear 0.25 MN/m (5.85 Hz).
- **Brakes** bias 0.50 → 0.58, 15 kN m default.
- **Footprint** 5.006 × 2.07 m (FIA 2017 overhang, width, wheelbase).
- I_z kept at 1159.6 (variant `iz560`).

Porpoising

- Stiff heave springs with zero heave damping leave a lightly damped **4.5 Hz** mode.
- The aero map adds 2 % downforce per cm the car drops, so it grows with v^2.
- Front ride-height std at 90 m/s: **9.8 mm** vs 3.0 mm before; not numerical (same at 0.5 ms). Heave damping 0.3 cuts it ≈10×.

Effect of `sf70h_f1`: lap time and controller

Lap [s], ERS off	`sf70h_tum`	`sf70h_f1`
QSS Shanghai, safe line	103.56	102.40
QSS Barcelona, safe line	82.89	82.44
OCP Shanghai	94.25	94.62
OCP Barcelona	78.76	79.38
Closed-loop Shanghai	102.87	101.84

Peak grip and understeer gradient are unchanged. The OCP is slower because the car is wider (narrower corridor).

Closed loop: TC target vs tyre

TC slip target	Lap 2 [s]
0.08 (old; = new Fx peak)	spin
0.04 / 0.05	103.9 / 102.7
0.06 / 0.07	**101.8** / spin
0.06 + heave damping 0.3	101.29

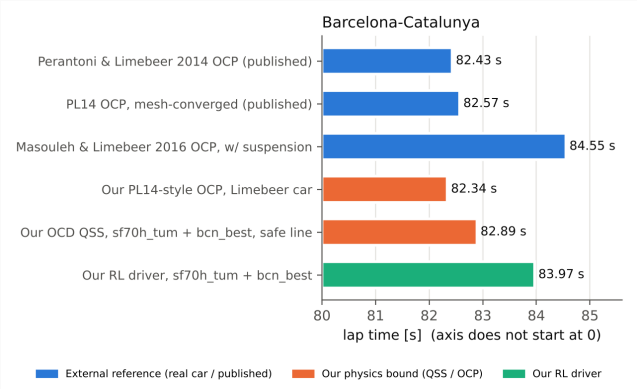
The TC target alone is worth $\approx$2 s. For RL: scale env steering to 0.30 rad and re-tune the demo driver's TC.

Verification: Limebeer-style minimum-lap-time OCP

- Planar body, quasi-static load transfer, 4 combined-slip tyres, LSD, aero, power limit. Direct collocation in distance (CasADi + IPOPT), mesh down to 0.5 m.
- **Barcelona with Limebeer's car: 82.34 s** vs PL14 mesh-converged 82.57 s: $-0.28\,\%$ (pass band $\pm 1.5\,\%$).
- Mesh-converged within 0.005 s; re-integration agrees to < 1 ms.
- Same model with SF70H-TUM values: **Shanghai 94.25 s** (TUM QSS 94.89 s), Barcelona 78.76 s.

Limits: local optima only (another initial guess gave $+0.55$ s); no suspension or wheel-spin dynamics. An external reference, not an OCD lap.

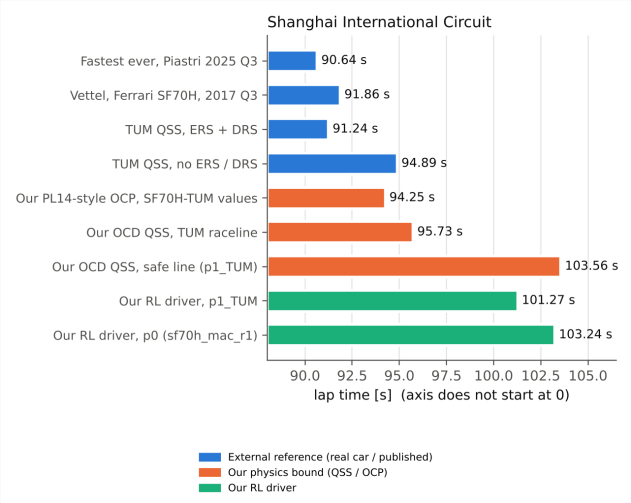
Barcelona: Limebeer's OCP vs. our RL driver



- RL: **83.97 s** (step 5.05M of 7.05M).
- +1.40 s (+1.7 %) to PL14's 82.57 s.
- +1.08 s (+1.3 %) to our own QSS (82.89 s).
- Faster than ML16's suspension model (84.55 s).

Not the same car: PL14 is a 2013-era F1 car (660 kg, 560 kW). Ours is `sf70h_tum` (733 kg, 575 kW) with `bcn_best` at aero_level 1.0 on a PLACEHOLDER aero map.

Shanghai: real world vs. QSS vs. our RL driver



No Limebeer lap exists for Shanghai.

- RL **101.27 s**: +9.4 s to Vettel's SF70H (91.86 s), +6.4 s to TUM QSS without ERS/DRS.
- The gap to the real car went from 31.0 s (AV-21, old reward) to 9.4 s.
- RL beats our QSS on its own safe line (103.56 s).

Summary and next steps

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- Time-optimal reward (no jitter term, linear $|e|$ as PBRS): the driver beats the QSS lap on its own reference line.
- The AV-21 tyre slip stiffness explains much of the gap to an F1 car; `sf70h_f1` replaces it with F1-sourced values.
- Our OCP reproduces Limebeer's Barcelona lap within 0.3 %; RL is within 1.7 % of it (different car).

Decisions needed

1. Heave damping for `sf70h_f1`, or softer heave springs (porpoising).
2. I_z: keep 1159.6 (recommended) or use 560 kg m^2.
3. RL on `sf70h_f1`: env steering to 0.30 rad, re-tuned demo TC. Then ERS on, a racing line sized for the footprint, seeds 1–2.

Still AV-21 / PLACEHOLDER: I_x, I_y, wheel masses, combined-slip weights, relaxation lengths, roll centres, dampers, aero map.